Entry Costs and Increasing Trade

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Abstract

Using confidential microdata from the US Census, we find that the fraction of manufacturing plants that export rose from 21% in 1987 to 39% in 2006. It has been suggested that similar trends in other countries may have been caused by declining costs of entering foreign markets. Our study tests this hypothesis for the first time. Both reduced form and structural estimation approaches find little evidence that entry costs declined significantly for US firms over this period. Despite the large literature on changes in variable costs to trade such as tariffs, our estimations represent the first analysis of how the costs of entering foreign markets have changed over time.

* Any opinions and conclusions expressed herein are those of the authors and do not neces sarily represent the views of the U.S. Census Bureau, the Board of Governors of the Federal Reserve System, or any other person associated with the Federal Reserve System. All results have been reviewed to ensure that no confidential information is disclosed. We especially thank our dissertation chairs Andrei Levchenko and Jim Levinsohn. Dan Ackerberg, Vanessa Alviarez, Wenjie Chen, Alan Deardorff, Ying Fan, Aaron Flaaen, Rob Feenstra, Jeremy Fox, Fariha Kamal, Bill Kerr, Brian Kovak, Pravin Krishna, C.J. Krizan, Pawel Krolikowski, Rosana Lincoln, Day Manoli, Prachi Mishra, Ryan Monarch, Dan Murphy, Justin Pierce, Mark Roberts, Mine Senses, Jagadeesh Sivadasan, Heiwai Tang, James Tybout, and Jing Zhang have all provided helpful comments. The staff at the U.S. Census Bureau has been exceptionally helpful, particularly Clint Carter, Al Cruz, James Davis, Barbara Downs, Cheryl Grim, Shawn Klimek, Margaret Levenstein, Danielle Sandler, and William Wisniewski. We thank Alex Avramov, Hannah Kwon, Alejandro Perez-Segura, and Isaac Rabbani for superb research assistance. The authors can be reached at william.lincoln@cmc.edu and andrew.h.mccallum@frb.gov. All errors are our own.

1 Introduction

A common feature of the rise in aggregate exports from several countries across the world is a significant expansion in the number of firms that export. A natural explanation that has been suggested by prior authors (e.g., Melitz 2003) is that the upfront costs of entering foreign markets have declined.[1] We test this idea for the first time using plant level data from the United States Census. We find that the US also saw significant foreign market entry over the period, with the fraction of plants that export rising from 21% in 1987 to 39% in 2006.[2] Across a number of different estimation approaches, however, we find little evidence for the idea that declines in the costs of entering foreign markets played a significant role in driving these trends. Our findings represent the first attempt to quantify how barriers to entry in foreign markets have changed over time and we hope our analysis will contribute to a better understanding of the sources of the rise in exporting.

Our analysis begins by presenting a number of descriptive statistics that provide new insight into the US experience. We find that the rise in the fraction of plants selling abroad mentioned above was broad-based; it was experienced across a range of industries as well as geographic regions. These extensive margin adjustments were matched with strong intensive margin adjustments, with average foreign sales per exporter also increasing substantially. Over time, changes along both of these margins had a large influence on aggregate trade volumes.

We next turn to understanding how much of a role declines in the costs of entering foreign markets played in these trends. As these costs cannot be directly observed with current data sources, we need to use models of firm behavior to estimate their magnitude. Thus, to get a comprehensive perspective we consider several different approaches. Our regression analyses provide a tractable way of addressing this question for the US manufacturing sector as a whole and allow for a wide variety of robustness checks. This approach, however, does not allow us to directly estimate the magnitude of changes in these costs. In our estimations, coefficient parameters in the regression specification are directly related to the costs of entering foreign markets. We let these coefficients differ across the earlier and later parts of the sample to look at how the costs compare. Our estimates imply similar magnitudes for these parameters across the two different time periods. These findings suggest small changes in the barriers to entry in foreign markets.

We then turn to a set of structural estimations that use the methodology developed by Das, Roberts, and Tybout (2007). This approach allows us to estimate the average level of foreign market entry costs that plants face in a given period. The methodology is attractive in that it provides numerical estimates of how these costs have changed and is flexible in accounting for other factors that determine exporting behavior. Estimations require the use of computationally

[1] See also Roberts and Tybout (1997a).

[2] We discuss our data and how these and other figures are calculated in Section 2.

intensive Bayesian Monte Carlo Markov Chain methods, however. We are thus constrained to focusing our analysis on understanding the experiences of a small set of industries. We estimate these costs across 1987-1997 and 1995-2006 and compare the results for the two time periods. For one industry we find increases in these costs, for a second we find declines, and for a third we find declines for smaller plants and increases for larger plants. Taken together, the results from the regression and structural estimations are evidence that declines in the costs of entering foreign markets have been modest at best and are unlikely to have played a large role in the changes that we see in the data.

Our work addresses an issue that is relevant for a number of other countries in addition to the US. Several other studies have suggested that large-scale foreign market entry was experienced worldwide during this period. Indeed, of the studies that have used plant or firm level data to study the rise in exports from other nations, many have found that entry into foreign markets played a significant role in the expansion of trade. This work includes studies on the experiences of Chile, Colombia, Costa Rica, India, Mexico, and Morocco.[3] We also see dramatic increases in the number of different goods sold across countries in disaggregated industry-level trade data. These results are consistent with substantial foreign market entry by firms in different sectors for a wide range of countries. Papers documenting these trends include Evenett and Venables (2002), Broda and Weinstein (2006), and Harris, Kónya, and Mátyás (2011). Particularly notable is an acceleration in the growth of varieties traded during our sample period of 1987-2006. Taken together, these studies suggest that our estimations address a question of first-order importance for understanding the recent growth of worldwide trade.

Our analysis also fills a significant gap in the international trade literature. A large number of studies have looked at the effect of changes in variable trade costs on export and import patterns. While there has been some work on other factors such as transportation costs, this work has primarily focused on understanding the effects of changes in tariffs. Yet these costs are only one, albeit important, piece of the puzzle. Changes in the barriers to entry in foreign markets also can have significant effects on trade patterns. One reason why these changes have not yet been studied is that methods to estimate their magnitude have only been developed relatively recently. Another is that the data requirements for looking at how they have changed are quite high. This study represents an initial effort to address this issue.

In the next section, we discuss our data sources and document several new stylized facts about US plants' exporting behavior from 1987 to 2006. Section 3 uses a model of export behavior to motivate regression based estimations on the evolving nature of these costs. In Section 4 we describe the structural model

[3] These papers include Roberts and Tybout 1997a, Mukerji 2009, Bergoeing, Micco, and Repetto 2011, and Lederman, Rodríguez-Clare, and Xu 2011. In the US context, Bernard and Jensen (2004a) have also previously documented a significant increase in the fraction of manufacturing plants that export over the period 1987-1992. Bernard, Jensen, and Schott (2009) additionally report significant extensive margin entry for US firms in goods (agriculture, manufacturing, and mining) sectors across the two years 1993 and 2000.

that we use to estimate changes in these costs and the results that we get from our estimations. Section 5 concludes.

2 Data and Stylized Facts

We use data from a number of different sources. Information on price deflators is obtained from the NBER manufacturing productivity database (Bartelsman and Gray, 1996). The primary microdata for our analyses come from the Annual Survey of Manufactures (ASM) and Census of Manufactures (CMF) from the United States Census Bureau. Both data sets contain information on the operations of US manufacturing plants. The CMF is conducted every year ending in 2 or 7 (e.g. 1987, 1992, etc.) and contains data on the universe of manufacturing establishments. The ASM is a survey of plants that is conducted in each intervening year. The sampling frames for these surveys are chosen two years after the most recent CMF. These establishments are then followed over time for five years until the next ASM sampling frame is implemented. Given the inability to aggregate to the firm level in the ASM, we treat the plant as the unit of analysis. This is consistent with the literature that has used this data as well as a number of other trade-related studies on other countries. We begin our analyses in 1987, the first year that comprehensive data on export revenues was collected.

The sample designs of these data sets impose some structure on our analysis. The ASM includes large plants with certainty but samples smaller plants according to their contribution to output. Due to the loss of non-certainty cases across different ASM panels, we limit our sample for panel analyses to plants with 250 or more employees. This avoids a number of challenges involved in following smaller plants over time and allows for comparability with previous studies that have used a similar approach. Despite this restriction, however, our data covers a significant portion of economic activity and the great majority of export volume.[4]

With these data we document a number of new stylized facts regarding the pace and character of trade growth since 1987. Figure 1 plots the percentage of plants with 20 or more employees that export in each year from 1987 to 2006.[5]

[4] Over the period 1987-1998 plants with more than 250 employees were sampled with certainty in the ASM. In the 1999-2003 ASM this threshold was increased to 500 employees and it was further raised to 1000 in the 2004-2008 ASM. As the sampling probability is inversely related to a plant's contribution to output, plants between 250 and 500 employees are still sampled with a high degree of certainty after 1998, however. In our estimations that span these years, we reweight the plants accordingly. Bernard and Jensen (2004a) use a similar sample and note that it accounts for 41% of employment, 52% of shipments, and 70% of exports in 1987.

[5] Similar to several other studies, in Figure 1 we focus on plants with 20 or more employees. Due to disclosure concerns, estimates for 1987 and 1992 are from Bernard and Jensen (2004b). In all of our analyses we drop administrative records, which are essentially imputed data for small employers and new businesses.

The overall upward trend is unmistakable; 21% of plants exported in 1987 and 39% exported in 2006. A number of different aspects of these trends are of note. First, we can get a sense of how much of these trends were due to adjustments in exporting status by existing establishments. Amongst plants that had 20 or more employees in both the 1987 and 2002 Census of Manufactures, 29% export in 1987 and 39% export in 2002. These figures suggest that a large part of these trends were due to adjustments by plants that were in operation in 1987 but only sold domestically. Secondly, taking the 21% participation rate from 1987 as a baseline, new plants that entered the sample and remained in business until 2002 were somewhat more apt to sell abroad. Those that exited were only slightly less likely to be exporters. The difference between these two figures consequently added to the overall trend but was not the sole determining factor. These trends and foreign market entry by existing plants both contributed.

The trends depicted in Figure 1 are also essentially the same when we consider plants with 10 or more or 250 or more employees, with the exception that the percentages are systematically shifted down or up in each year. This reflects the fact that smaller plants are significantly more likely to only sell domestically. These results are especially important for our estimation approach in the next section in which we are limited to plants with at least 250 employees, in that same basic patterns in the data are similar for smaller plants. Indeed, we find the same basic trends in export participation when looking separately across categories of the employment size distribution. We additionally find significant increases in exporting at the level of the firm in Census of Manufactures years (1987 and 2002), for which we can aggregate the operations of plants.

Table 1 and Figures 2 and 3 look at the sectoral and geographic dimensions of the rise in export market participation. Figure 2 plots the percentage of plants that export in each industry in 1987 and 2006. While some industries saw larger changes than others, there has been a significant expansion in foreign market participation across nearly all sectors of the economy. Figure 3 similarly demonstrates that the results in Figure 1 were experienced broadly across different regions of the US. These results generally hold across states as well. In Tables 2 and 3 we document the time path of each of these trends. While we find similar patterns to the overall trend by region, there is more heterogeneity in the timing and magnitude of foreign market entry across industries. The fact that the expansion in the fraction of plants that export has been pervasive across these two dimensions suggests that these trends were not driven by idiosyncratic factors such as the rise of high-tech industries.

In a similar vein, we looked at how the rise of exporting varied across plants of different sizes. The intuition developed from a number of models suggests that these trends may have been driven by smaller producers (e.g. Melitz 2003, Helpman, Melitz, and Yeaple 2004, and Chaney 2008). We find that this was in fact not the case. In Figure 4 we look at the change in the probability that a plant exports across different employment size categories. While there is some variation in the increase across different types of plants, the rise of exporting was experienced by each group. This suggests that the welfare effects from these changes were likely larger than they would have been if they were driven solely

by smaller plants.

To get a sense of how changes in the extensive margin have affected overall trade volumes, we use information from the years in which we have data from the Census of Manufactures. This allows us to track the universe of small as well as large plants over time. The fact that the intensive margin dominates trade volumes in the short-run has been documented by, among others, di Giovanni and Levchenko (2009) and Bernard, Jensen, Redding, and Schott (2007b). Authors have only recently begun to focus on the relative importance of the extensive margin for aggregate trade volumes over longer time horizons, however. Table 5 reports the contribution to Census year aggregate exports by plants that exported in a given prior Census year. Only 46% of aggregate exports in 2002 came from plants that exported in 1987, 1992, and 1997. These numbers underestimate the importance of changes along the extensive margin since they are not restricted to plants that exported continuously in all prior years.[6]

3 Regression Evidence

In this section we consider regression evidence on how the costs of entering foreign markets have changed over time. While our structural estimations in the next section will allow us to study different industries in depth, this analysis will give us a sense of how these costs have changed for the manufacturing sector as a whole. Drawing upon the seminal work of Dixit (1989) and Baldwin and Krugman (1989), several prior studies have used a simple binary choice model of whether or not to export to test for the existence of barriers to entry in foreign markets.[7] Here, we use this approach to get a sense of how these costs have changed over time. The basic premise of the model is that a plant will sell abroad if the benefits from exporting exceed the additional costs of doing so. The benefits include the extra gross profits that it could make as well as any option value associated with being an exporter in the future. The costs include barriers to entry for plants that did not export previously. Specifically, a plant that has not exported for more than two years must pay a sunk cost F_0 to enter the foreign market and a re-entry cost F_R if it last exported two years ago. The model leads to a simple decision rule where

[6] We are unable to calculate year-to-year statistics based on continuously exporting plants due to the breaks between ASM panels. These figures echo related results reported in Bergoeing, Micco, and Repetto (2011) for Chile 1990-2007, Bernard, Jensen, Redding and Schott (2007b) for the aggregate US economy (including non-manufacturing sectors) for 1993-2003, and Eaton, Eslava, Kugler and Tybout (2007) for Colombia 1996-2005.

[7] See Roberts and Tybout (1997b), Bernard and Wagner (2001), and Bernard and Jensen (2004a). For related work, see also Alessandria and Choi (2007), Arkolakis (2010), Moxnes (2010), Hanson and Xiang (2011), Aw, Roberts, and Xu (2011), Dutt, Santacreu, and Traca (2015), McCallum (2015), Dickstein and Morales (2015), and Morales, Sheu, and Zahler (2015).

$$y_{it} = \begin{cases} 1 & \text{if } p^*_{it} - F_0 + F_0 \cdot y_{it-1} + (F_0 - F_R) \cdot \tilde{y}_{it-2} \geq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Here y_{it} is plant $i's$ export status in year t and $\tilde{y}_{it-2} = y_{it-2}(1 - y_{it-1})$ is an indicator function for whether the plant last exported two years prior to year t. The term p^*_{it} can be written as

$$p^*_{it} = p_{it} + \delta \left(E_t \left[V_{it+1} \mid y_{it} = 1 \right] - E_t \left[V_{it+1} \mid y_{it} = 0 \right] \right).$$

It is determined by the extra gross profit that the plant could make by exporting this year p_{it} plus the option value associated with being an exporter next period. This option value, in turn, is given by the difference in the discounted future expected value of being an exporter today relative to only selling domestically.

If there are no costs to entering the foreign market, the condition for exporting in equation (1) collapses to $p_{it} \geq 0$. In this case, the plant decides whether or not to export based solely on what is most profitable today and ignores dynamic considerations. Thus, once controlling for factors that account for changes in p_{it}, if there are no costs to entering the foreign market we should see a lack of state dependence in exporting status.

To obtain an estimating equation that will allow us to look at changes in F_0 and F_R we need to parameterize $p^*_{it} - F_0$. A number of factors likely influence this term, such as changes in plant productivity and fluctuations in foreign income. We use the following functional form

$$p^*_{it} - F_0 \approx \mu_i + X'_{it}\beta + \phi_t + \varepsilon_{it}$$

to develop the specification

$$y_{it} = \mu_i + X'_{it}\beta + \alpha_1 \cdot y_{it-1} + \alpha_2 \cdot \tilde{y}_{it-2} + \phi_t + \varepsilon_{it}. \quad (2)$$

This equation provides the basis for our estimations. The vector X_{it} contains a number of covariates that predict export market participation. These include the ratio of nonproduction to total employment and the logarithms of employment, total factor productivity, and average wages. We also include an industry-level trade-weighted exchange rate series.[8] Productivity is estimated with the approach of Levinsohn and Petrin (2003). Since information on exports in the ASM is only available beginning in 1987 and our approach includes a covariate based on twice lagged export status, we begin the analysis in 1989 and use 1987 and 1988 as pre-sample years. Unobserved plant specific factors that influence p^*_{it} are captured in the term μ_i. Business cycle effects and other time varying factors are absorbed into the year fixed effects ϕ_t. The coefficients $\alpha_1 = F_0$ and $\alpha_2 = (F_0 - F_R)$ parameterize the importance of barriers to entry

[8] Each exchange rate is a geometric export-weighted average of bilateral real exchange rates where the weights are constructed using 3 digit SIC export data. We follow the aggregation method used by the US Federal Reserve, as detailed in Loretan (2005). We use the same industry-level exchange rate series for the estimations in both Sections 3 and 4.

in foreign markets. Larger estimates of α_1, for example, suggest higher sunk costs F_0.[9]

In column (1) of Table 6 we present our findings from estimating equation (2) as presented earlier. Standard errors in parentheses are clustered at the plant level. Controlling for other factors, exporting last year raises a plant's probability of exporting this year by 44 percent. These results are consistent with the prior literature on testing for the existence of barriers to entry. In column (2) we include interaction terms of the variables y_{it-1} and $\tilde{y}_{it-2}$ with an indicator function for the post-1998 period $Post_{98}$. The coefficient estimates on these interaction terms are given by how the costs F_0 and F_R compare in the second half of the sample with those in the first. We find little change in the coefficient α_1 in the second part of the panel and a somewhat larger decrease in α_2. This suggests relatively small changes in the costs F_0 and an increase in the sunk costs of re-entering foreign markets F_R. In column (3) we additionally include interactions of the variables in X_{it} with $Post_{98}$ and the results do not change significantly.

We come to similar conclusions when considering alternative approaches. These include using different years for the post-period (e.g. $Post_{99}$), only considering plants with 350 or more employees, using different covariates in X_{it}, estimating productivity with other approaches, and adding the variable "Last exported three years ago" and its interaction with $Post_{98}$. Simple estimations of the specification in (2) without fixed effects or plant level controls also suggest little change in export status persistence.

In these baseline estimations, we make no restrictions on entry into or exit out of the sample. We also do not find large declines in entry costs when using a balanced panel and these results are robust to the alternative approaches described earlier. This is reassuring not only for the validity of our results here but also for our subsequent estimations in which we are constrained to use a balanced panel. In a similar vein, we performed estimations like those described earlier but restricted the sample to the industries that we considered for our structural estimations in the next section. We come to similar conclusions here as well, which suggests that the industries that we chose to focus on are broadly representative of overall trends.

[9] Prior studies have found little difference between the costs of entering foreign markets anew and those of entering after two years of not exporting. They have also found a small difference between F_0 and F_R above. We find similar results. The model can be extended to include a cost of leaving the foreign market L, which makes the coefficient α_1 in equation (2) a function of $F_0 + L$. It can also be changed to allow for the upfront costs of learning about the foreign market, where $p^*_{it} - F_0$ is a function of prior export status y_{it-1}. This change similarly makes the coefficient on y_{it-1} in equation (2) a function of both the costs of learning and the costs F_0.

4 Structural Estimation

4.1 Model

In this section, we turn to a different approach to address how the costs of entering foreign markets have evolved. The extra structure afforded by the model here allows us to provide numerical estimates of the costs of entering foreign markets in different time periods. Specifically, we use the estimation methodology developed by Das, Roberts, and Tybout (2007) to look at the average level of foreign market entry costs facing plants over the 1987-1997 and 1995-2006 periods. Comparing these cost estimates across the two panels will then give us a sense of how they have changed over time. In addition to addressing the question of the determinants of the rise in export intensity, our results contribute to the emerging literature on estimating the magnitude of these barriers. Indeed, these are the first estimates of these costs for the United States.

Here we lay out the basics of the model underlying the estimation approach; further details are contained in the appendix. All plants in the model serve the domestic market and face the choice of whether or not to sell their goods abroad. The foreign and domestic markets are segmented from one another and are both monopolistically competitive. We abstract from entry and exit into production in the domestic market, requiring the use of a balanced panel in our estimations. We assume that plants' marginal costs do not respond to output shocks, simplifying the model significantly by isolating the decision to serve foreign markets from domestic concerns. Plants are forward-looking in the sense that, although they do not know what their future realizations of marginal costs, foreign demand, and the exchange rate will be, they know the Markov processes by which these factors evolve and set their expectations accordingly.

The log of potential profits from selling in the foreign market π^*_{it} for plant i in year t is defined as

$$\ln(\pi^*_{it}) = \psi_0 z_i + \psi_1 e_t + v_{it} \tag{3}$$

where z_i indexes time-invariant plant characteristics and e_t is the exchange rate facing the plant. v_{it} is a stationary, serially correlated disturbance term that captures shifts in factors that determine potential export profits. Examples of these factors include changes in productivity, tariffs, and demand. Although this general form is quite parsimonious, it allows for significant flexibility in accounting for many of the other potential explanations for changes in export status. We assume that v_{it} is the sum of m stationary and independent $AR(1)$ processes. Formally, we have $v_{it} = \sum_{j=1}^{m} x_{jit}$ where i indexes plants, t the time period, and j the type of potential shock. Each of these potential shocks can be written $x_{jit} = \lambda^j_x x_{jit} + w_{xjt}$, where w_{xjt} is normally distributed with mean zero and variance σ^2_{wj}. The composite term v_{it} therefore follows an $ARMA(m, m-1)$ process. The exchange rate e_t follows the AR(1) process

$e_t = \lambda_0 + \lambda_e e_{t-1} + w_{et}$ where w_{et} is normally distributed with mean zero and variance σ_w^2. The parameters λ_0, λ_e, σ_w and the distribution of w_{et} are known to all plants. For ease of exposition, we denote $\Psi = (\psi_{01}, ..., \psi_{0k}, \psi_1) = (\psi_0, \psi_1)$ and collect the parameters λ_x^j and σ_{wj} into the diagonal matrices Λ_x and $\sum_\omega$.

The relevant variable for the empirical analysis of a plant's decision of whether or not to export is the level of foreign profits that it could make. Our data, however, only contain information on total revenues and export revenues. In order to make estimation possible we draw upon two aspects of the model mentioned above: first, markets are monopolistically competitive, and second, foreign and domestic markets are segmented. We further denote c_{it} as the marginal cost of production, $\eta_i > 1$ as a plant-specific foreign demand elasticity, and P_{it}^f as the domestic currency price of exports. If the plant exports, it would optimally choose to price its goods such that $c_{it}^f = P_{it}^f \left(1 - \eta_i^{-1}\right)$. This implies that potential foreign revenues R_{it}^{f*} and variable costs C_{it}^{f*} to exporting can be written as $C_{it}^{f*} = R_{it}^{f*} \left(1 - \eta_i^{-1}\right)$ if we multiply both sides of this expression by the optimal quantity of exports. Using the fact that $\pi_{it}^* = R_{it}^{f*} - C_{it}^{f*}$, this condition implies that potential export profits are given by

$$\pi_{it}^* = \eta_i^{-1} R_{it}^{f^*}. \tag{4}$$

Taking logs and substituting this expression into (3) yields

$$\ln\left(R_{it}^{f^*}\right) = \ln\left(\eta_i\right) + \psi_0 z_i + \psi_1 e_t + v_{it}. \tag{5}$$

This relationship provides a way to estimate the parameters that determine export profits and allows us to account for a significant amount of plant heterogeneity. It does, however, create an incidental parameters problem with the introduction of the terms $\eta = \{\eta_i\}_{i=1}^n$. As the number of plants in the sample grows, so too does the number of parameters.

To identify η we explicitly use data on costs and revenues. We begin by assuming that the ratio of foreign demand elasticities to domestic demand elasticities is $1 + \upsilon$ for all plants in the industry. By steps analogous to those used to derive (4), profit maximization and segmented markets imply that we should observe $C_{it}^d = R_{it}^d \left(1 - \eta_i^{-1}[1 + \upsilon]\right)$ in the domestic market. Combining this with (4) and invoking the assumption of segmented markets, optimally selected production for all markets must satisfy

$$C_{it} = C_{it}^f + C_{it}^d = R_{it}^f \left(1 - \eta_i^{-1}\right) + R_{it}^d \left(1 - \eta_i^{-1}\left(1 + \upsilon\right)\right). \tag{6}$$

Dividing this expression by $R_{it} = R_{it}^f + R_{it}^d$, rearranging, replacing optimal with realized values, and including an error term ξ_{it} yields

$$1 - \frac{C_{it}}{R_{it}} = \eta_i^{-1}\left(1 + \upsilon \frac{R_{it}^d}{R_{it}}\right) + \xi_{it}. \tag{7}$$

Here R_{it}^d, R_{it}, and C_{it} are the plant's realized domestic revenue, total revenue, and total variable cost. We assume that the error term ξ_{it} comes from measurement error in the costs C_{it} and follows the AR(1) process $\xi_{it} = \lambda_\xi \xi_{it-1} + w_{\varsigma t}$,

where $w_{\varsigma t}$ is normally distributed with variance σ_{ς}^2. We can then use this expression to form the density function $f_c\left(C_{i0}^T \mid R_{i0}^{fT}, R_{i0}^{dT}, \theta\right)$.

Equation (2) gives us an expression for the baseline level of profits that plants earn from foreign markets in each period. In looking at the plant's dynamic problem of whether or not to export, we further allow each plant to receive a shock to profits each period of $\kappa + \varepsilon_{1it}$. κ is common to all plants and ε_{1it} is allowed to vary across plants i and years t. Plants must also pay an upfront, sunk cost to enter foreign markets $\gamma_s(z_i) + \varepsilon_{2it} - \varepsilon_{1it}$. These one-time costs γ_s depend on time invariant plant characteristics z_i, are paid fully in the first year of exporting, and are allowed to vary across plants and time. Examples of these costs include market research, setting up distribution channels, learning about foreign regulations and documentation requirements, and a number of other non-tariff barriers. It is the estimation of these parameters γ_s in which we are most interested. Note that γ_s parameterizes the typical costs that plants face and not necessarily the costs that are paid by plants that begin to sell abroad. Indeed, all else equal, the plants that enter are those that are likely to have drawn a favorable shock of $\varepsilon_{2it} - \varepsilon_{1it}$. We assume that ε_{jit} are serially uncorrelated, normally distributed with mean zero and variance $\sigma_{\varepsilon j}^2$, and are uncorrelated with v_{it} and e_t. For the sake of exposition, we let $\sum_{\varepsilon} = diag(\varepsilon_{1it}, \varepsilon_{2it})$ and $\Gamma = (\gamma_{s1}, \gamma_{s2}, ..., \gamma_{sk}, \kappa) = (\gamma_s, \kappa)$.

We are now in a position to describe the plant's decision of whether or not to export. Let y_{it} be an indicator variable for whether plant i exported in year t. Using the expression for gross potential export profits π_{it}^* from (3), we can write

$$u(\cdot) = \begin{cases} \pi_{it}^*(e_t, x_{it}, z_i) + \kappa + \varepsilon_{1it} & \text{if } y_{it} = 1 \text{ and } y_{it-1} = 1 \\ \pi_{it}^*(e_t, x_{it}, z_i) + \kappa - \gamma_s(z_i) + \varepsilon_{2it} & \text{if } y_{it} = 1 \text{ and } y_{it-1} = 0 \\ 0 & \text{if } y_{it} = 0. \end{cases} \tag{8}$$

The plant's potential net export profits depend on its prior export status, since we assume that sunk costs have to be paid if the plant did not export in the previous year.

In each period t, the plant observes the values of e_t, x_{it}, ε_{jit}, and z_i and forms its expectations about the future using the fact that it knows the processes by which these terms evolve. The plant then determines the decision rule of whether or not to export $y_{it} = y(e_t, x_{it}, z_i, \varepsilon_{jit}, y_{it-1} \mid \theta)$ which maximizes its net discounted expected profit stream over a 30 year horizon. Formally, we have the Bellman equation

$$V_{it} = \max_{y_{it} \in \{0,1\}} \{u(e_t, x_{it}, z_i, \varepsilon_{jit}, y_{it-1}, y_{it} \mid \theta) + \delta E_t V_{it+1}\} \tag{9}$$

where

$$E_t V_{it+1} = \int_{e'} \int_{x'} \int_{\varepsilon'} V_{it+1} \cdot f_e(e' \mid e_t, \theta) \cdot f_x(x' \mid x_t, \theta) \cdot f_\varepsilon(\varepsilon' \mid \varepsilon_t, \theta) \, d\varepsilon' dx' de'$$

and θ collects all the parameters

$$\theta = (\Psi, \eta, \upsilon, \Lambda_x, \Sigma_\omega, \Gamma, \Sigma_\varepsilon, \lambda_0, \lambda_e, \sigma_w, \lambda_\xi, \sigma_\varsigma).$$

The decision rule of whether or not to export can be written as a binary choice problem $y_{it} = I(y_{it}^* > 0)$. Here $I(\cdot)$ is an indicator function and y_{it}^* is a comparison of the benefits from exporting and from not exporting

$$y_{it}^* = u(e_t, x_{it}, z_i, \varepsilon_{it}, 1, y_{it-1} \mid \theta) + \delta \Delta E_t V_{it+1}(e_t, x_{it}, z_i \mid \theta) \tag{10}$$

where

$$\Delta E_t V_{it+1}(e_t, x_{it}, z_i \mid \theta) = E_t[V_{it+1} \mid y_{it} = 1] - E_t[V_{it+1} \mid y_{it} = 0].$$

The first term in (10) reflects the direct benefits today from exporting, whereas the second term reflects the option value of being an exporter tomorrow.

4.2 Estimation

Using the expressions developed above to describe a plant's intensive and extensive margin exporting decisions, we then develop a likelihood function that allows us to estimate the parameters in one step

$$L(D \mid \theta) = \prod_{i=1}^{n} f_c\left(C_{i0}^T \mid R_{i0}^{fT}, R_{i0}^{dT}, \theta\right) \cdot P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i, \theta\right). \tag{11}$$

Here $D = \{D_i\}_{i=1}^n$ denotes the data for all plants. $f_c\left(C_{i0}^T \mid R_{i0}^{fT}, R_{i0}^{dT}, \theta\right)$ is determined by the expression in (7) and the likelihood $P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i, \theta\right)$ is formed from the relationships implied by the extensive margin decision. We provide more details about the construction of $P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i, \theta\right)$ in the appendix.

Estimating the likelihood function $L(D \mid \theta)$ with classical methods presents two problems. First, while this feature of the approach allows us to account for a significant amount of plant heterogeneity, we are faced with an incidental parameters problem in that we need to estimate $\eta = \{\eta_i\}_{i=1}^n$. To add to this, the likelihood function is highly non-standard and unlikely to be globally concave in θ. To circumvent these issues, we use a Bayesian approach and write the posterior distribution of the parameters with $P(\theta \mid D) \propto q(\theta) L(D \mid \theta)$, where $q(\theta)$ gives our prior beliefs about the parameters. To characterize the posterior distribution $P(\theta \mid D)$, we then use the random walk Metropolis-Hastings algorithm. This algorithm essentially allows us to estimate $E(\theta \mid D)$ by performing Monte Carlo integration using a Markov chain.

Computational constraints place some restrictions on the level of heterogeneity for which these estimates can account. To characterize the time invariant plant characteristics that affect sunk costs and export profits, we let z_i equal an indicator function based on plant size. The threshold for z_i is set to be equal to the median level of sales in 1987, such that half of the plants are considered large in the first panel for each industry. We keep this threshold for the second panel, capturing changes in plant sales. The number of $AR(1)$ processes additively included in the profit function disturbance term is set to two $v_{it} = x_{1it} + x_{2it}$, intuitively reflecting separate cost and demand shock processes. We set the discount rate δ to 0.9. In order to ease computational costs, we do not estimate the parameters for the exchange rate process simultaneously with the rest of the model. Instead, we estimate them separately using export-weighted industry real exchange rates constructed with the same approach as those described in Section 3. We fit each of these series with an $AR(1)$ process from 1972 until the last year of each panel to give estimates of $\hat{\lambda}_0$, $\hat{\lambda}_e$, and $\hat{\sigma}_w$. These parameters are then treated as fixed for the purposes of the estimation of the model.

For the rest of our parameters, we have to specify a prior distribution. With a few exceptions, we make these distributions reasonably diffuse to let the data speak for itself. To impose non-negativity on the variance parameters, our priors are that they are distributed log normally with a mean of zero and a variance of 2. Our priors on the root of each AR(1) process are that they are distributed uniformly on $(-1, 1)$. This ensures that these processes are stationary. We use a more restrictive prior for η_i. Following the empirical literature, we set the prior such that $\ln(\eta_i - 1) \sim N(2, 1)$. This implies a mean and standard deviation for η_i of 12.2 and 16.0, respectively. It also ensures that $\eta_i > 1$, which is a necessary condition for the model. The prior for υ, the parameter that determines the ratio of foreign and domestic demand elasticities, is also assumed to be uniform on $[-5, 5]$. The priors for other parameters are given in Table 7.

Given these preliminaries, it is possible to provide intuition about the main sources of variation used to identify the sunk cost parameters. First note that for any type of plant the probability of exporting is an increasing function of the gross potential profit stream that it could earn in foreign markets. If there are no barriers to entry, the probability that a plant exports today should not depend on whether it exported yesterday. Plants with similar gross potential profit streams should have the same probability of exporting regardless of their exporting history. If there are significant upfront costs, however, plants that previously exported should have a higher probability of exporting ceteris paribus since they do not need to pay to enter. The higher these costs are, the bigger the difference should be between plants that exported previously and those that did not. Thus, differences in the exporting frequencies of plants with similar gross potential export profit streams but different exporting histories in our data provide significant identifying variance for the sunk cost parameters.

4.3 Results

The assumptions of the model make it appropriate to consider different industries separately. At the same time, one of the primary limitations of the estimation approach is that it is highly computationally intensive. In choosing which industries to focus on, we used several criteria to narrow down our choices: (i) there were enough plants in each panel to allow for identification, (ii) the industry was sufficiently export oriented, (iii) like the manufacturing sector as a whole, the overall destination composition of industry exports was relatively stable from 1987 to 2006, and (iv) in order to get a broad view, the industries were in different two digit SIC sectors. These criteria led us to consider three particular three digit SIC industries: Preserved Fruits and Vegetables (SIC 203), Aircraft and Parts (SIC 372), and Measuring and Controlling Devices (SIC 382). Table 8 presents the four digit industries that constitute each of these three digit industries.

Tables 9-11 present results for each industry. For each parameter we report the estimated mean and standard deviation and all figures are in 1987 dollars. Following the recommended strategy for posterior simulation suggested by Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin (2003), to construct our estimates we consider 100,000 post-burn in draws from the posterior distribution from three separate chains. Our estimates are thus each based on a total of 300,000 draws. We allow for a 50,000 draw burn in period for each chain and the convergence of each chain is checked using the diagnostic tests reviewed in Brooks and Roberts (1998). Despite generally using highly diffuse priors, the posterior distributions for most of our parameters are significantly concentrated. This suggests that the estimates are primarily informed by the data rather than our choice of priors.

Consistent with the results from the previous section, we find generally comparable results for the sunk cost parameters γ_s across the two different time periods. For Preserved Fruits and Vegetables we find modest declines across both plant size categories, for Aircraft and Parts we find increases across both categories, and for Measuring and Controlling Devices we find a decline for smaller plants and an increase for larger plants. Elasticity estimates are also consistent with the values suggested by the literature. Calculations using the elasticity estimates for each plant suggest that the magnitude of the sunk costs is typically roughly equal to a few years of the average level of exporting profits. The sunk cost parameters for each industry are generally between 2 and 3 million dollars and the standard deviations of the posterior distributions range from 360 thousand dollars to 1.21 million dollars. Thus, while these posterior distributions are not concentrated enough to make stark claims on the precise change in barriers to entry, in concert with our results from Section 3 they do suggest that large scale declines are unlikely.

In interpreting these results more broadly, a number of factors likely worked to increase as well as decrease these costs. For example, trade agreements could have lowered them. At the same time, in what little survey evidence we have,

firms list market research and redesigning their products for foreign markets as two of the primary costs that they face in beginning to sell abroad. With the increasing integration of the world economy, market research costs may have increased substantially due to the need to identify and study competition from a greatly expanded number of source countries. Additionally, there is evidence that technical barriers to trade have increased over time, particularly among countries that have had their hands tied with respect to tariffs due to international agreements. Many of these policy changes likely raised the costs of entry.[10] We consider the effects of these factors to be an open area for future research.

5 Conclusion

In this study we have documented a significant shift towards exporting for US plants over 1987-2006. In looking at why this occurred we considered a natural explanation that has been suggested as a potential cause for similar trends in other countries: declines in the upfront costs of entering foreign markets. Across different approaches to understanding this issue, we show that reductions in these barriers were unlikely to have played a significant role in these trends. Our work represents an initial attempt to understand how the costs of entering foreign markets have evolved over time.

We close with a discussion of a few areas of research that are likely to be fruitful for future work. Firstly, qualitative evidence on the determinants of export market entry costs would be tremendously valuable. Despite the evidence presented here and their ubiquity in trade models, there is surprisingly little direct survey evidence about these costs. Retrospective research in this area could help us better understand the results presented above. Secondly, much of the work on understanding the effects of free trade agreements focuses on how declines in tariffs affect aggregate trade volumes. Total trade tends to increase through extensive margin adjustments following these agreements, however, and the details of these accords often include provisions likely to reduce barriers to entry. Disentangling these effects would significantly improve our understanding of how different impediments affect trade and would likely yield more accurate analyses of potential policy changes. Finally, an improved understanding of the experiences of other countries would also provide further insight into the evolution of the barriers to entry in foreign markets. We attempted to obtain data to expand our analysis to countries beyond the US, but were unable to locate a data set with sufficient history and detail. Further analyses on the

[10] For evidence on changes in technical barriers to trade, see Baldwin (2000), Maskus, Wilson, Otsuki (2000), U.S. Department of Commerce (2004), United Nations Conference on Trade and Development (2005), and U.S. Trade Representative (2011). For survey evidence on the nature of barriers to entry, see the study conducted for the World Bank found in First Washington Associates (1991).

experiences of firms in other countries would add greatly to our understanding of the worldwide rise in international trade.

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7 Appendix

In this appendix we provide further details about our structural estimation approach. We begin by describing how we develop the extensive margin likelihood function and the density functions for foreign market profit shocks. We then describe our approach to calculating the option value associated with exporting $\Delta E_t V_{it+1}\left(e_t, z_i, x_{it} \mid \theta\right)$. A description of our Bayesian MCMC estimation approach closes. The discussion of the model here and in the main text follows Das, Roberts, Tybout (2007); see this paper for further details about the model and estimation approach.

7.1 Extensive Margin Likelihood

For the purposes of estimation, we can connect the binary choice decision problem laid out in the body of the text to a likelihood function that uses our data from US plants. We begin by writing observed export profit shocks as

$$v_i^+ = \left\{ \ln\left(R_{it}^f\right) - \ln\left(\eta_i\right) - \psi_0 \cdot z_i - \psi_1 \cdot e_t \mid R_{it}^f > 0 \right\}.$$

We can then write the export profit shock for plant i in each year t as a function of these observed shocks and a set of m *iid* standard normal random variates μ_i such that $x_{it} = x_{it}\left(v_i^+, \mu_i\right)$. For each plant, we can write

$$\begin{aligned} P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i\right) &= P\left(y_{i0}^T, v_i^+ \mid e_0^T, z_i\right) \\ &= P\left(y_{i0}^T \mid e_0^T, z_i, v_i^+\right) \cdot h\left(v_i^+\right) \\ &= \left[\int_{\mu_i} P\left(y_{i0}^T \mid e_0^T, z_i, x_0^T\left(v_i^+, \mu_i\right)\right) \cdot g\left(\mu_i\right) d\mu_i\right] \cdot h\left(v_i^+\right) \end{aligned}$$

where the density functions for μ_i and v_i^+ are given by $g\left(\mu_i\right)$ and $h\left(v_i^+\right)$. We discuss how to construct $g\left(\mu_i\right)$, $h\left(v_i^+\right)$ and the term $\Delta E_t V_{it+1}\left(e_t, x_{it}, z_i \mid \theta\right)$ in the next sections of the appendix. The value of $P\left(y_{i0}^T \mid e_0^T, z_i, v_i^+\right)$ will be calculated using the distribution of $g\left(\mu_i\right)$ and Monte Carlo integration, drawing several μ_i from $g\left(\mu_i\right)$, plugging into $P\left(y_{i0}^T \mid e_0^T, z_i, x_0^T\left(v_i^+, \mu_i\right)\right)$, and averaging. The term $P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i, \theta\right)$ can then be linked to our data by factoring out the initial conditions such that

$$\begin{aligned} P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i\right) &= P\left(y_{i1}^T \mid e_1^T, z_i, x_1^T\left(v_i^+, \mu_i\right), y_{i0}\right) \cdot \\ &\quad P\left(y_{i0} \mid e_0, z_i, x_0\left(v_i^+, \mu_i\right)\right) \end{aligned}$$

We use Heckman's (1981) solution to the initial conditions problem, and estimate $P\left(y_{i0} \mid e_0, z_i, x_0\left(v_i^+, \mu_i\right)\right)$ using

$$\begin{aligned} P\left(y_{i0} \mid e_0, z_i, x_0\left(v_i^+, \mu_i\right)\right) &= \left(\Phi\left(\alpha_0 + \alpha_1' z_i + \alpha_2' x_0\left(v_i^+, \mu_i\right)\right)\right)^{y_{i0}} \cdot \\ &\quad \left(1 - \Phi\left(\alpha_0 + \alpha_1' z_i + \alpha_2' x_0\left(v_i^+, \mu_i\right)\right)\right)^{1-y_{i0}} \end{aligned}$$

Using backward induction along with Rust's (1997) random grid algorithm, we can calculate $\Delta E_t V_{it+1}\left(e_t, x_{it}, z_i \mid \theta\right)$ in each period. We then further use the export market participation rule in (8) to develop the likelihood function

$$P\left(y_{i1}^T \mid e_1^T, z_i, x_1^T\left(v_i^+, \mu_i\right), y_{i0}\right) =$$

$$\begin{aligned} &\prod_{i=1}^T \left[E_{\varepsilon_{it}}\left(I\left(y_{it}^* > 0 \mid e_t, z_i, x_t\left(v_i^+, \mu_i\right), \varepsilon_{it}, y_{it-1}\right)\right)\right]^{y_{it}} \cdot \\ &\left[E_{\varepsilon_{it}}\left(I\left(y_{it}^* \leq 0 \mid e_t, z_i, x_t\left(v_i^+, \mu_i\right), \varepsilon_{it}, y_{it-1}\right)\right)\right]^{1-y_{it}} \end{aligned}$$

Differences across plants and time in terms of export market participation, costs, and foreign and domestic sales will then help pin down our parameters of interest. In particular, variation in export market participation by plants that would earn similar levels of profits in export markets but that are different in terms of their prior foreign market presence will be important in identifying sunk entry costs.

7.2 Density Functions for Foreign Market Profit Shocks

In this section we describe how we construct $h\left(v_i^+\right)$ and $x_0^T\left(v_i^+, \mu_i\right)$. These are elements that form part of $P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i\right)$. We begin by deriving the density function for

$$\begin{aligned} v_i^+ &= \left\{\ln\left(R_{it}^f\right) - \ln\left(\eta_i\right) - \psi_0 \cdot z_i - \psi_1 \cdot e_t \mid R_{it}^f > 0\right\} \\ &= \left\{v_{it} \equiv \iota' x_{it} \mid R_{it}^f > 0\right\}. \end{aligned}$$

For each plant we observe $q_i = \sum_{t=0}^T y_{it}$ values of v_i^+. We first assume that each x_{it} process is in long-run equilibrium such that $x_{it} \sim N\left(0, \Sigma_\omega\left(I - \Lambda_x^2\right)^{-1}\right)$. Thus, we have $h\left(v_i^+\right) = N\left(0, \Sigma_{vv}\right)$ where $E\left[v_{it}^2\right] = \iota'\left(x_{it} x_{it}'\right)\iota = \iota' \Sigma_\omega\left(I - \Lambda_x^2\right)^{-1}\iota$ and $E\left[v_{it} v_{it-k}\right] = \iota' \Lambda_x^{|k|} \Sigma_\omega\left(I - \Lambda_x^2\right)^{-1}\iota$ where $k \neq 0$.

The next key element in constructing $P\left(y_{i0}^T, R_{i0}^{fT} \mid e_0^T, z_i\right)$ is to develop the function $x_0^T\left(v_i^+, \mu_i\right)$. We first write x_{i0}^T as an $mT \times 1$ vector $x_{i0}^T = \left(x_{i0}', \ldots, x_{iT}'\right)'$. Given the $q_i \times 1$ vector v_i^+ we can write

$$x_{i0}^T \mid v_i^+ \sim N\left(\Sigma_{xv}\Sigma_{vv}^{-1}v_i^+, \Sigma_{xx} - \Sigma_{xv}\Sigma_{vv}^{-1}\Sigma_{xv}'\right).$$

Here $\Sigma_{xx} \equiv E\left(x_{i0}^T \cdot x_{i0}^{T\ \prime}\right)$ and $\Sigma_{xv} \equiv E\left(x_{i0}^T \cdot v_i^{+\ \prime}\right)$; the elements of these matrices are given by $E\left(x_{it} \cdot x_{it+s}'\right) = \Lambda_x^{|s|} \cdot \Sigma_\omega \cdot \left(I - \Lambda_x^2\right)^{-1}$ and $E\left(x_{it} \cdot v_{it+s}\right) = \Lambda_x^{|s|} \cdot \Sigma_\omega \cdot \left(I - \Lambda_x^2\right)^{-1} \iota$. See Chow (1983) for further discussion.

We can then use these expressions to write

$$x_{i0}^T = x_{i0}^T\left(v_i^+, \mu_i\right) = \begin{cases} Av_i^+ + B\mu_i & \text{if } q_i > 0 \\ B\mu_i & \text{if } q_i = 0. \end{cases}$$

Here $A = \Sigma_{xv}\Sigma_{vv}^{-1}$, $BB = \Sigma_{xx} - \Sigma_{xv}\Sigma_{vv}^{-1}\Sigma_{xv}'$, and μ_i is an $mT \times 1$ vector of iid standard normal random variables with density function $g\left(\mu_i\right) = \prod_{j=1}^{mT} \phi\left(\mu_{ij}\right)$. We can use this expression to form $x_{it} = x_t\left(v_i^+, \mu_i\right)$ and $x_{is}^T = x_s^T\left(v_i^+, \mu_i\right)$ that are then a part of

$$P\left(y_{i0}^T \mid e_0^T, z_i, v_i^+\right) = \int_{\mu_i} P\left(y_{i0}^T \mid e_0^T, z_i, x_{i0}^T\left(v_i^+, \mu_i\right)\right) \cdot g\left(\mu_i\right) \cdot d\mu_i.$$

Specifically, we can then use this functional form to simulate $P\left(y_{i0}^T \mid e_0^T, z_i, v_i^+\right)$. This is done by (i) drawing a set of S vectors μ_i from $g\left(\mu_i\right)$ (ii) using the values to calculate $x_{i0}^T\left(v_i^+, \mu_i\right)$ and (iii) averaging over the resulting values to calculate $P\left(y_{i0}^T \mid e_0^T, z_i, v_i^+\right)$.

7.3 Calculating the Option Value $\Delta E_t V_{it+1}\left(e_t, z_i, x_{it} \mid \theta\right)$

In obtaining an estimate of the latent value of exporting

$$y_{it}^* = \left[u\left(e_t, z_i, x_{it}, \varepsilon_{it}, y_{it} = 1, y_{it-1} \mid \theta\right) - 0\right] + \delta\Delta E_t V_{it+1}\left(e_t, z_i, x_{it} \mid \theta\right)$$

the term $u\left(e_t, z_i, x_{it}, \varepsilon_{it}, y_{it} = 1, y_{it-1} \mid \theta\right)$ can be calculated using the functional forms presented in the text. To obtain an estimate for $\Delta E_t V_{it+1}\left(e_t, z_i, x_{it} \mid \theta\right)$ we begin by using backward induction over a 30 year time horizon to first calculate

$$V_{it}^O = \delta E_t V_{it+1}\left(e_{t+1}, x_{it+1}, z_i \mid y_{it} = 0, \theta\right)$$
$$V_{it}^E = \pi\left(e_{t+1}, x_{it+1}, z_i, \theta\right) - \kappa - \gamma_s \cdot z_i + \delta E_t V_{it+1}\left(e_{t+1}, x_{it+1}, z_i \mid y_{it} = 1, \theta\right)$$
$$V_{it}^S = \pi\left(e_{t+1}, x_{it+1}, z_i, \theta\right) - \kappa + \delta E_t V_{it+1}\left(e_{t+1}, x_{it+1}, z_i \mid y_{it} = 1, \theta\right).$$

Here V_{it}^O is the expected value of only selling domestically in period t, V_{it}^E is the expected value from entering the foreign market, and V_{it}^S is the expected value

of continuing to sell abroad. The algorithm begins in the last year in which $E_t V_{it+1} = 0$ and then calculates V_{it}^O, V_{it}^E, and V_{it}^S backwards successively until the current period is reached. We use Rust's (1997) random grid algorithm to integrate numerically over the state variables x and e. We calculate

$$\begin{aligned} E_t\left[V_{it+1} \mid y_{it}=1\right] &= E_t \max\left(V_{it+1}^O, V_{it+1}^S + \varepsilon_{1it+1}\right) \\ &= \int_{x_{t+1}} \int_{e_{t+1}} \left[\begin{array}{c} \Phi\left(\frac{V_{it+1}^S - V_{it+1}^O}{\sigma_{\varepsilon 1}}\right) \times \left[V_{it+1}^S + \sigma_{\varepsilon 1} \cdot \left[\frac{\phi\left(\frac{V_{it+1}^S - V_{it+1}^O}{\sigma_{\varepsilon 1}}\right)}{\Phi\left(\frac{V_{it+1}^S - V_{it+1}^O}{\sigma_{\varepsilon 1}}\right)} \right] \right] \\ + \Phi\left(\frac{V_{it+1}^O - V_{it+1}^S}{\sigma_{\varepsilon 1}}\right) \cdot V_{it+1}^O \end{array} \right] \\ &\quad \cdot f\left(x_{t+1} \mid x_t\right) \cdot f\left(e_{t+1} \mid e_t\right) \cdot dx_{t+1} \cdot de_{t+1} \end{aligned}$$

and

$$\begin{aligned} E_t\left[V_{it+1} \mid y_{it}=0\right] &= E_t\left[\max\left(V_{it+1}^O, V_{it}^E + \varepsilon_{2it+1}\right)\right] \\ &= \int_{x_{t+1}} \int_{e_{t+1}} \left[\begin{array}{c} \Phi\left(\frac{V_{it}^E - V_{it+1}^O}{\sigma_{\varepsilon 2}}\right) \cdot \left[V_{it}^E + \sigma_{\varepsilon 2} \cdot \left[\frac{\phi\left(\frac{V_{it}^E - V_{it+1}^O}{\sigma_{\varepsilon 2}}\right)}{\Phi\left(\frac{V_{it}^E - V_{it+1}^O}{\sigma_{\varepsilon 2}}\right)} \right] \right] \\ + \Phi\left(\frac{V_{it+1}^O - V_{it}^E}{\sigma_{\varepsilon 2}}\right) \cdot V_{0it+1} \end{array} \right] \\ &\quad \cdot f\left(x_{t+1} \mid x_t\right) \cdot f\left(e_{t+1} \mid e_t\right) \cdot dx_{t+1} \cdot de_{t+1}. \end{aligned}$$

7.4 Monte Carlo Markov Chain Methods

We take $S = 100k$ draws of the posterior distribution $P\left(\theta \mid D\right)$ from three separate chains to construct our estimates using the random-walk Metropolis-Hastings algorithm. These draws are taken after an initial burn-in period of $50k$ draws for each chain that allows the chain to converge to the posterior distribution. The means and standard deviations are estimated with $\bar{\theta} = \frac{1}{S}\sum_{s=1}^{S} \theta^s$ and

$$\sqrt{\frac{1}{S}\sum_{s=1}^{S}\left(\theta^s - \bar{\theta}\right)\cdot\left(\theta^s - \bar{\theta}\right)'}$$

where θ^s is a given draw of the entire parameter vector from the posterior distribution. We use a Metropolis-Hastings algorithm in which we update the different components of the parameter vector separately in each iteration of the chain. We choose to partition θ with $\theta^s = (\theta_1^s, \theta_2^s, \ldots, \theta_8^s)$ where $\theta_1 = \Psi$, $\theta_2 = \Lambda_x$, $\theta_3 = \Sigma_\omega$, $\theta_4 = \Gamma$, $\theta_5 = \Sigma_\varepsilon$, $\theta_6 = \eta$, $\theta_7 = (\upsilon, \rho, \sigma_\xi)$, $\theta_8 = \varsigma$. Once starting values for the chain are chosen, for each iteration we perform the following steps. These steps are then repeated for each iteration.

1. Draw a potential new value for one of the subvectors θ_i based on the value from the previous iteration of the chain. This can be written as $\tilde{\theta}_i^* = \tilde{\theta}_i^s + \upsilon_i^s$ where $\tilde{\theta}_i^s$ is the value of the subvector from the previous iteration and υ_i^s is a mean-zero vector of shocks. The covariance matrix for υ_i^s, Σ_{υ_i}, is chosen before the estimations begin and is held fixed throughout.

2. Define $\tilde{\theta}_{-i}^s$ as the set of parameters in θ excluding those in $\tilde{\theta}_i^s$. Calculate the ratio

$$\alpha_i^s = min\left(\frac{P\left(\theta_i^* \mid \theta_{-i}^s, D\right)}{P\left(\theta_i^s \mid \theta_{-i}^s, D\right)}, 1\right)$$

and update the set of parameters θ_i with

$$(\theta_i^{s+1}, \theta_{-i}^s) = \begin{cases} \left(\theta_i^*, \theta_{-i}^s\right) & \text{with probability } \alpha_i^s \\ \left(\theta_i^s, \theta_{-i}^s\right) & \text{with probability } 1 - \alpha_i^s. \end{cases}$$

3. Conduct the same process for each block of parameters θ_i. Once this is done $\forall\ i$, we take the resulting value of θ as our draw from the chain. This process is repeated for each draw of the chain.

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Table 1: Industry, Regional, and Plant Size Decomposition

Industry	1987	2006	Region	1987	2006
Food	15	30	New England	25	46
Beverage & Tobacco	45	30	Middle Atlantic	19	39
Textile Mill Products	16		East North Central	25	45
(Textile Mills)		47	West North Central	23	41
(Textile Product Mills)		35	South Atlantic	18	36
Apparel	5	19	East South Central	18	36
Wood products	12	19	West South Central	19	33
Furniture	10	20	Mountain	18	30
Paper	19	43	Pacific	21	36
Printing & Publishing	5	15			
Chemicals	40	63			
Petroleum & Coal	22	37	Employment	1987	2006
Plastics & Rubber	26	51	20-49	11	23
Leather	19	47	50-149	22	41
Non-metallic Minerals	14	17	150-249	33	54
Primary Metals	27	56	250-499	42	61
Fabricated Metals	21	32	500-999	54	69
Machinery	33	62	1000+	70	78
Electrical Equipment	37	65			
Instruments/Computer & Electronic Products	48	68			
Transportation Equipment	29	56			
Misc. Manufacturing	20	42			

Notes: The table lists the percentage of manufacturing plants that export using the Census of Manufactures in 1987 and the Annual Survey of Manufactures in 2006. Results are broken down by industry, region, and firm size. The states corresponding to each region are listed in Table 4. Due to concerns about disclosure, the industry figures for 1987 are from Bernard and Jensen (2004b) and we report the results for 1987 in two digit 1987 SIC codes and the results for 2006 in three digit 2002 NAICS codes. These industry codes match well at this level of aggregation. Estimations are for plants with 20 or more employees. Increases in exporting are found across industries, regions, and plant size categories.

Table 2: Plant Export Participation by Industry

	Percent of Plants That Export				
Industry	1987	1992	1997	2003	2006
Food	15	23	25	27	30
Tobacco	45	51	47		
(Beverage & Tobacco)				28	30
Textile Mill Products	16	25	28		
(Textile Mills)				40	47
(Textile Product Mills)				30	35
Apparel	5	9	13	13	19
Wood Products	12	18	16	16	19
Furniture	10	25	24	18	20
Paper	19	31	32	35	43
Printing & Publishing	5	10	11	14	15
Chemicals	40	49	49	55	63
Petroleum & Coal	22	30	30	31	37
Plastics & Rubber	26	36	39	40	51
Leather	19	28	35	38	47
Non-metallic Minerals	14	21	20	17	17
Primary Metals	27	39	39	43	56
Fabricated Metals	21	31	32	30	32
Machinery	33	43	41	56	62
Electronic & Other Electric Equipment	37	46	47		
(Electrical Equipment, etc.)				54	65
Instruments	48	55	56		
(Computer & Electronic Products)				58	68
Transportation Equipment	29	40	41	49	56
Miscellaneous Manufacturing	20	34	36	37	42
Total	21	30	32	35	39

Notes: The table lists the percentage of manufacturing plants that export in each industry using the Census of Manufactures in 1987, 1992, and 1997 as well as the Annual Survey of Manufactures in 2003 and 2006. Due to concerns about disclosure, the results reported for 1987 and 1992 are from Bernard and Jensen (2004b) and we report the results for 1987 in two digit 1987 SIC codes and the results for 2003 and 2006 in three digit 2002 NAICS codes. These codes match well at this level of aggregation. As in Bernard and Jensen (2004b), estimates for all years are for plants with 20 or more employees. The figures suggest that there was significant heterogeneity across industries in the time path of the rise of exporting.

Table 3: Plant Export Participation by Region

	Percent of Plants That Export				
Region	1987	1992	1997	2003	2006
New England	25	37	37	42	46
Middle Atlantic	19	29	30	34	39
East North Central	25	34	35	39	45
West North Central	23	32	33	37	41
South Atlantic	18	27	29	32	36
East South Central	18	27	27	30	36
West South Central	19	28	28	31	33
Mountain	18	26	27	32	30
Pacific	21	31	31	33	36
Total	21	30	32	35	39

Notes: The table lists the percentage of manufacturing plants that export in each U.S. Census division using the Census of Manufactures in 1987, 1992, and 1997 as well as the Annual Survey of Manufactures in 2003 and 2006. We report the states corresponding to these divisions in Table 4. Estimates for all years are for plants with 20 or more employees. These results suggest that the time path of participation rates of each region roughly matches the overall trend across these years.

Table 4: Census Division of the States

Census Division	State	Census Division	State
New England	Connecticut	East South Central	Alabama
	Maine		Kentucky
	Massachusetts		Mississippi
	New Hampshire		Tennessee
	Rhode Island		
	Vermont	West South Central	Arkansas
			Louisiana
Middle Atlantic	New Jersey		Oklahoma
	New York		Texas
	Pennsylvania		
		Mountain	Arizona
East North Central	Indiana		Colorado
	Illinois		Idaho
	Michigan		New Mexico
	Ohio		Montana
	Wisconsin		Utah
			Nevada
West North Central	Iowa		Wyoming
	Nebraska		
	Kansas	Pacific	Alaska
	North Dakota		California
	Minnesota		Hawaii
	South Dakota		Oregon
	Missouri		Washington
South Atlantic	Delaware		
	District of Columbia		
	Florida		
	Georgia		
	Maryland		
	North Carolina		
	South Carolina		
	Virginia		
	West Virginia		

Notes: The table lists the states corresponding to each of the Census Divisions.

Table 5: Intensive Margin

	Starting			
Continuing	1987	1992	1997	2002
1987	1			
1992	0.75	1		
1997	0.58	0.79	1	
2002	0.46	0.58	0.71	1

Notes: The table lists the percentage of exports in each Census of Manufactures (CMF) year that came from plants that exported in each of the previous Census years, starting in 1987. Thus only 46% of exports in 2002 came from plants that exported in 1987, 1992, and 1997. Similar to our other figures, estimations are limited to plants with 20 or more employees.

Table 6: Fixed Effects Estimations

	Specification		
	Without Interactions	Baseline	Additional Interactions
Variable	(1)	(2)	(3)
Exported last year	0.439**	0.436**	0.431**
	(0.005)	(0.006)	(0.006)
Exported last year * $Post_{98}$		0.005	0.012**
		(0.005)	(0.006)
Last exported two years ago	0.103**	0.140**	0.138**
	(0.007)	(0.009)	(0.009)
Last exported two years ago * $Post_{98}$		-0.091**	-0.087**
		(0.013)	(0.013)
ln (Employment)	0.031**	0.031**	0.030**
	(0.005)	(0.005)	(0.006)
ln (Wages)	0.026**	0.026**	0.039**
	(0.008)	(0.008)	(0.010)
Non-production/Total Employment	-0.021	-0.021	-0.003
	(0.016)	(0.016)	(0.018)
ln (Productivity)	0.005**	0.005**	0.007**
	(0.002)	(0.002)	(0.002)
ln (Exchange Rate)	0.060	0.057	0.031
	(0.031)	(0.032)	(0.044)
Plant Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Interactions between X_{it} and $Post_{98}$	No	No	Yes
R^2	0.552	0.553	0.552

Notes: The table presents the results from the fixed effects estimations. The baseline estimations in column (2) suggest no decline of initial entry costs and an increase in re-entry cost. The coefficient on "Exported last year" is an increasing function of the costs of entering foreign markets anew, F_0. The coefficient on "Last exported two years ago" is similarly an increasing function of the difference $F_0 - F_R$, where F_R is the cost of re-entering foreign markets after leaving in the previous year. $Post_{98}$ is an indicator function for the post-1998 part of the sample. The dependent variable is a 0/1 indicator for a given plant's export status in the current year. Column (1) presents the results with no interactions and column (2) contains our baseline results. Column (3) reports results from additionally including interactions between the variables in X_{it} and $Post_{98}$. Plant specific characteristics in X_{it} are lagged by one period in all specifications. All estimations include 106,000 observations (this figure is rounded to the nearest 100 observations for the purposes of disclosure). Standard errors in parentheses are clustered at the plant level. The results with ** denote significance at the 5 percent level.

Table 7: Prior Distributions

	Profits
ψ_{01} (intercept)	$\psi_{01} \sim N(0, 10)$
ψ_{02} (dom. size dummy)	$\psi_{02} \sim N(0, 10)$
ψ_1 (exchange rate)	$\psi_1 \sim N(0, 10)$
λ_x^1 (root, first AR)	$\lambda_x^1 \sim U(-1, 1)$
λ_x^2 (root, second AR)	$\lambda_x^2 \sim U(-1, 1)$
$\sigma_{\omega 1}^2$ (variance, first AR)	$\ln(\sigma_{\omega 1}^2) \sim N(0, 20)$
$\sigma_{\omega 2}^2$ (variance, second AR)	$\ln(\sigma_{\omega 2}^2) \sim N(0, 20)$
υ (foreign elas. premium)	$\upsilon \sim U[-5, 5]$
λ_ξ (root, measurement error)	$\lambda_\xi \sim U(-1, 1)$
σ_ξ (std. dev., measurement error)	$\ln(\sigma_\xi) \sim N(0, 2)$
	Elasticities of Demand
η_i (demand elasticity)	$\ln(\eta_i - 1) \sim N(2, 1)$
	Exporting Decision
γ_{s1} (sunk cost, small plants)	$\gamma_{s1} \sim N(0, 20)$
γ_{s2} (sunk cost, large plants)	$\gamma_{s2} \sim N(0, 20)$
κ (mean, ε_1 & ε_2)	$\kappa \sim N(0, 20)$
$\sigma_{\varepsilon 1}$ (st. dev., ε_1)	$\ln(\sigma_{\varepsilon 1}) \sim N(0, 2)$
$\sigma_{\varepsilon 2}$ (st. dev., ε_2)	$\ln(\sigma_{\varepsilon 2}) \sim N(0, 2)$
	Initial Conditions
α_0 (intercept)	$\alpha_0 \sim N(0, 50)$
α_1 (dom. size dummy)	$\alpha_1 \sim N(0, 50)$
α_2 (x_1)	$\alpha_2 \sim N(0, 50)$
α_3 (x_2)	$\alpha_3 \sim N(0, 50)$

Notes: The table presents the priors used for our structural estimations. The same distributions are used for each panel and industry. We generally choose diffuse priors to allow the data to speak for themselves. Variance parameters have log normal distributions to impose non-negativity. The root of each $AR(1)$ process is bounded on $(-1, 1)$ in order to ensure stationarity. The notation here is the same as in the paper by Das, Roberts, and Tybout (2007) and the reader is referred to this work for additional details about the estimation methodology.

Table 8: Four Digit Subindustries for MCMC Estimations

3 Digit SIC Industry	4 Digit SIC Subindustry
Preserved Fruits and Vegetables (203)	Canned specialties (2032)
	Canned fruits and vegetables (2033)
	Dehydrated fruits, vegetables, and soups (2034)
	Pickles, sauces, and salad dressings (2035)
	Frozen fruits and vegetables (2037)
	Frozen specialties, N.E.C. (2038)
Aircraft and Parts (372)	Aircraft (3721)
	Aircraft Engines and Engine Parts (3724)
	Aircraft Parts and Equipment, N.E.C. (3728)
Measuring and Controlling Devices (382)	Laboratory Apparatus and Furniture (3821)
	Environmental Controls (3822)
	Process Control Instruments (3823)
	Fluid Meters and Counting Devices (3824)
	Instruments to Measure Electricity (3825)
	Analytical Instruments (3826)
	Optical Instruments and Lenses (3827)
	Measuring and Controlling Devices, N.E.C. (3829)

Notes: The table lists the four digit 1987 SIC industries that compose the three digit 1987 SIC industries that we consider for our MCMC estimations.

Table 9: SIC 203 Posterior Parameter Distributions

	Preserved Fruits & Vegetables	
	1987-1997	1995-2006
	Profits	
ψ_{01} (intercept)	-2.07 (0.21)	-2.08 (0.29)
ψ_{02} (dom. size dummy)	1.04 (0.28)	1.46 (0.38)
ψ_1 (exchange rate)	0.85 (1.49)	-0.70 (0.74)
λ_x^1 (root, first AR)	0.57 (0.08)	0.66 (0.10)
λ_x^2 (root, second AR)	0.64 (0.04)	0.91 (0.02)
$\sigma_{\omega 1}^2$ (variance, first AR)	0.68 (0.07)	0.27 (0.07)
$\sigma_{\omega 2}^2$ (variance, second AR)	0.73 (0.07)	0.43 (0.07)
υ (foreign elas. premium)	0.03 (0.04)	-0.13 (0.04)
λ_ξ (root, measurement error)	0.89 (0.01)	0.84 (0.02)
σ_ξ (std. error, measurement error)	0.23 (0.03)	0.18 (0.02)
	Elasticities of Demand	
η_μ (demand elas., μ across plants)	13.71 (8.81)	12.93 (6.16)
η_σ (demand elas., σ across plants)	11.61 (7.35)	12.59 (6.67)
	Exporting Decision	
γ_{s1} (sunk cost, small plants)	2.85 (0.80)	2.49 (0.41)
γ_{s2} (sunk cost, large plants)	2.62 (0.74)	2.26 (0.36)
κ (mean, ε_1 & ε_2)	-0.14 (0.04)	-0.17 (0.03)
$\sigma_{\varepsilon 1}$ (std. error, ε_1)	1.55 (0.48)	1.17 (0.32)
$\sigma_{\varepsilon 2}$ (std. error, ε_2)	0.90 (0.38)	0.62 (0.38)
	Initial Conditions	
α_0 (intercept)	8.68 (8.58)	-35.38 (16.49)
α_1 (dom. size dummy)	18.23 (20.73)	43.42 (20.58)
α_2 (x_1)	-23.95 (48.44)	17.41 (45.75)
α_3 (x_2)	9.33 (78.54)	-8.26 (61.25)
Observations	$N = 1200, T = 11$	$N = 1100, T = 12$

Notes: The table presents the results for the Preserved Fruits and Vegetables industry (SIC 203) over the time periods 1987-1997 and 1995-2006. All estimations are based on 100,000 draws from a post-burn in period from three separate chains, for 300,000 total draws. Means are presented along with standard deviations of the posterior distribution for each parameter in parentheses. Figures in dollars are in 1987 dollars. The number of observations in each panel has been rounded to the nearest 100 observations for the purposes of disclosure. Median values and estimates based on thinning each chain give similar results.

Table 10: SIC 372 Posterior Parameter Distributions

	Aircraft & Parts	
	1987-1997	1995-2006
	Profits	
ψ_{01} (intercept)	-0.36 (0.34)	-0.21 (0.31)
ψ_{02} (dom. size dummy)	2.62 (0.45)	2.81 (0.42)
ψ_1 (exchange rate)	-0.25 (0.97)	0.60 (0.57)
λ_x^1 (root, first AR)	0.31 (0.06)	0.80 (0.09)
λ_x^2 (root, second AR)	0.98 (0.01)	0.96 (0.02)
$\sigma_{\omega 1}^2$ (variance, first AR)	0.65 (0.06)	0.46 (0.10)
$\sigma_{\omega 2}^2$ (variance, second AR)	0.09 (0.04)	0.15 (0.10)
υ (foreign elas. premium)	2.03 (0.40)	1.42 (0.24)
λ_ξ (root, measurement error)	0.98 (0.01)	0.95 (0.01)
σ_ξ (std. error, measurement error)	1.29 (0.29)	0.74 (0.15)
	Elasticities of Demand	
η_μ (demand elas., μ across plants)	12.20 (4.82)	11.86 (4.17)
η_σ (demand elas., σ across plants)	12.80 (5.56)	12.16 (4.55)
	Exporting Decision	
γ_{s1} (sunk cost, small plants)	2.39 (0.62)	2.90 (0.65)
γ_{s2} (sunk cost, large plants)	2.45 (0.65)	2.54 (0.59)
κ (mean, ε_1 & ε_2)	-0.24 (0.07)	-0.26 (0.07)
$\sigma_{\varepsilon 1}$ (std. error, ε_1)	1.09 (0.40)	1.11 (0.29)
$\sigma_{\varepsilon 2}$ (std. error, ε_2)	0.85 (0.38)	1.06 (0.40)
	Initial Conditions	
α_0 (intercept)	45.99 (21.15)	24.97 (16.50)
α_1 (dom. size dummy)	8.04 (25.54)	35.75 (23.54)
α_2 (x_1)	-3.26 (25.49)	-8.34 (52.19)
α_3 (x_2)	38.58 (66.70)	56.54 (34.56)
Observations	$N = 900, T = 11$	$N = 1000, T = 12$

Notes: The table presents the results for the Aircraft and Parts industry (SIC 372) over the time periods 1987-1997 and 1995-2006. All estimations are based on 100,000 draws from a post-burn in period from three separate chains, for 300,000 total draws. Means are presented along with standard deviations of the posterior distribution for each parameter in parentheses. Figures in dollars are in 1987 dollars. The number of observations in each panel has been rounded to the nearest 100 observations for the purposes of disclosure. Median values and estimates based on thinning each chain give similar results.

Table 11: SIC 382 Posterior Parameter Distributions

	Measuring & Controlling Devices	
	1987-1997	1995-2006
	Profits	
ψ_{01} (intercept)	-0.18 (0.17)	0.26 (0.22)
ψ_{02} (dom. size dummy)	0.96 (0.24)	1.53 (0.31)
ψ_1 (exchange rate)	-0.95 (0.63)	0.09 (0.49)
λ_x^1 (root, first AR)	0.12 (0.26)	0.80 (0.20)
λ_x^2 (root, second AR)	0.91 (0.07)	0.92 (0.03)
$\sigma_{\omega 1}^2$ (variance, first AR)	0.19 (0.07)	0.16 (0.06)
$\sigma_{\omega 2}^2$ (variance, second AR)	0.16 (0.06)	0.09 (0.06)
υ (foreign elas. premium)	1.23 (0.16)	0.89 (0.26)
λ_ξ (root, measurement error)	0.97 (0.01)	0.95 (0.02)
σ_ξ (std. error, measurement error)	0.67 (0.11)	0.58 (0.18)
	Elasticities of Demand	
η_μ (demand elas., μ across plants)	10.95 (6.39)	10.58 (5.49)
η_σ (demand elas., σ across plants)	7.76 (4.81)	7.85 (4.46)
	Exporting Decision	
γ_{s1} (sunk cost, small plants)	2.89 (0.98)	2.62 (0.93)
γ_{s2} (sunk cost, large plants)	2.66 (0.89)	3.29 (1.21)
κ (mean, ε_1 & ε_2)	-0.56 (0.32)	-0.95 (0.74)
$\sigma_{\varepsilon 1}$ (std. error, ε_1)	1.60 (0.55)	1.35 (0.64)
$\sigma_{\varepsilon 2}$ (std. error, ε_2)	1.12 (0.84)	2.75 (2.17)
	Initial Conditions	
α_0 (intercept)	43.81 (19.79)	41.79 (24.86)
α_1 (dom. size dummy)	33.54 (23.61)	7.84 (21.88)
α_2 (x_1)	58.63 (26.13)	1.57 (56.36)
α_3 (x_2)	-13.26 (43.48)	41.30 (51.51)
Observations	$N = 1100, T = 11$	$N = 800, T = 12$

Notes: The table presents the results for the Measuring and Controlling Devices industry (SIC 382) over the time periods 1987-1997 and 1995-2006. All estimations are based on 100,000 draws from a post-burn in period from three separate chains, for 300,000 total draws. Means are presented along with standard deviations of the posterior distribution for each parameter in parentheses. Figures in dollars are in 1987 dollars. The number of observations in each panel has been rounded to the nearest 100 observations for the purposes of disclosure. Median values and estimates based on thinning each chain give similar results.